Written Justification

ARCHITECTURAL JUSTIFICATION

Student Project Portfolio & Showcase App

Architecture: Microservices Architecture

1. Architectural Choice

We chose **Microservices Architecture** for the **Student Project Portfolio & Showcase App**.

2. Reason 1 — Independent Functionality

The system contains several distinct functions such as **project showcase/profile management, multimedia artifact management, project evaluation, voting and voter verification, and leaderboard management**. These functionalities can be separated into independent services. This supports independent scaling, fault isolation, and independent deployment, which are characteristics of microservices architecture.

3. Reason 2 — Different Workloads

The application has different workloads. In particular, **public voting and leaderboard calculations** may receive many more requests than project profile management. Lab 1 also specifies that voting and leaderboard calculations should update in real time with an **API response latency under 300 ms**. Separating

these functions allows the high-traffic voting functionality to be scaled independently.

4. Security Advantage

Microservices allow security controls to be applied separately to sensitive functionality. For example, the **Voting and Voter Verification Service** can enforce voter verification and duplicate-vote prevention independently from the public showcase functionality. This addresses the Lab 1 requirement for strict **anti-sybil protection** and helps ensure that only authorized actions are performed.

5. Performance Benefit

The **Voting and Leaderboard Service** can be scaled independently when voting traffic increases. This allows the system to handle a large number of voting requests and maintain fast leaderboard updates without requiring the entire application to be scaled. This supports the Lab 1 requirement for real-time updates and an API response latency below 300 ms.